Sai Anil Kumar Deyyala

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Professional Summary

DevOps and Platform Engineer with 4 years and 8 months of experience designing, automating, and managing cloud infrastructure across Azure and AWS. Skilled in Kubernetes, AKS, Terraform, CI/CD, GitOps, infrastructure automation, observability, and cloud networking. Experienced in automating end-to-end infrastructure provisioning and microservices deployments using Terraform and CI/CD pipelines, building reusable deployment workflows, and implementing secure, scalable cloud-native platforms. Strong foundation in Machine Learning and MLOps, with exposure to AI infrastructure and cloud-native workloads.

Technical Skills

Operating Systems: Linux, Windows
Programming Languages: Python
Scripting: Bash, Python
Cloud Platforms: Microsoft Azure, AWS
Containerization: Docker
Container Orchestration: Kubernetes, Azure Kubernetes Service (AKS), Helm
Service Mesh & Ingress: Istio, NGINX Ingress
Infrastructure as Code (IaC): Terraform, Ansible
Infrastructure Testing & Security: Terratest, Checkov
CI/CD & GitOps: Azure DevOps, Jenkins, ArgoCD
Version Control: Git, GitHub
Monitoring & Observability: Prometheus, Grafana, Azure Monitor, Application Insights, Log Analytics
Cloud Networking & Security: VNet, NIC, NSG, Private Endpoints, Private Link Service, Load Balancer, Front Door, WAF, Key Vault, Managed Identity, Secrets Store CSI Driver
Machine Learning & MLOps: ML, ML Pipelines, MLflow, PyTorch, OpenCV, pandas, NumPy, Transfer Learning, Jupyter Notebook

Experience

Infrastructure Consultant July 2024 – Present
Thoughtworks Technologies (India) Pvt. Ltd. *Hyderabad, India*
Singapore Client – Central Government Project

- Design and maintain reusable GitLab CI/CD pipeline templates to automate infrastructure provisioning and application deployments across Azure environments.
- Deploy and operate microservices on Azure Kubernetes Service (AKS), developing Helm charts and configuring NGINX and Istio ingress for application traffic management.
- Integrate Kubernetes application secrets with Azure Key Vault using the Secrets Store CSI Driver and configure application connectivity to Azure database and platform services.
- Provision Azure infrastructure including MySQL and PostgreSQL Flexible Servers, Service Bus, Storage Accounts, Azure Functions, and Cosmos DB using automated deployment workflows.
- Plan and configure Azure networking through VNet address allocation, VNet peering, firewall rules, Private Endpoints, Azure Load Balancer, and federated identity integration.
- Configure Application Insights, Log Analytics, Azure Alerts, and dashboards to support application observability, troubleshooting, and operational monitoring.
- Perform Azure MySQL database and Storage Account migrations, estimate environment costs, and provision complete Azure environments.
- Collaborate with developers to troubleshoot application, networking, database, and deployment issues across cloud environments.

US Client – Azure Cloud Infrastructure Project

- Manage 15+ independent monolithic services hosted on Azure App Service and Azure Container Apps, supporting application infrastructure and deployment workflows.
- Develop reusable Bicep Infrastructure as Code (IaC) templates and Azure DevOps CI/CD pipelines to automate infrastructure provisioning and application deployments.

- Provision and manage Azure networking, compute, storage, database, security, and monitoring resources, including VNets, NSGs, Azure MySQL, SQL Database, Key Vault, Service Bus, Front Door, and Application Insights.
- Configure Managed Identities, Private Link Service, Private Endpoints, WAF, and Azure Load Balancer to support secure cloud infrastructure.
- Manage Azure Data Factory, Azure Functions, Storage Accounts, Container Registry, Virtual Machines, and Virtual Machine Scale Sets across application environments.

Senior DevOps Engineer June 2023 – June 2024
Tiger Analytics *India*

- Deployed a microservices application end to end on Azure Kubernetes Service (AKS), containerizing individual applications with Docker and orchestrating workloads through Kubernetes.
- Developed Kubernetes manifests for Deployments, Secrets, ConfigMaps, Services, and other application resources to standardize microservices deployments.
- Configured Istio ingress and service mesh resources, troubleshooting Kubernetes networking and application connectivity issues.
- Provisioned Azure infrastructure using reusable Terraform configurations and developed Azure DevOps CI/CD pipelines to automate infrastructure and application deployments.
- Implemented Prometheus and Grafana monitoring for Kubernetes nodes, cluster CPU and memory utilization, and service-level metrics including request traffic and throughput.
- Delivered a chaos engineering proof of concept to explore application resilience and infrastructure failure scenarios.
- Migrated Azure Data Factory and Azure Databricks workloads between environments, supporting consistent data platform deployments.

DevOps Engineer August 2022 – May 2023
Tiger Analytics *India*

- Deployed and managed Azure and AWS infrastructure using Terraform, integrating Azure DevOps and Jenkins pipelines for automated provisioning.
- Built CI/CD pipelines for infrastructure and application deployments, including deployment automation for a monolithic Java application.
- Migrated Azure Data Factory and Azure Databricks assets across environments to support consistent data platform configurations.
- Integrated Terratest for infrastructure testing and Checkov for infrastructure-as-code policy scanning to improve deployment validation and security checks.
- Developed automation scripts using Python, Go, and Shell scripting to support infrastructure operations and deployment workflows.
- Worked with Docker, Kubernetes, Linux, Git, GitHub, Prometheus, and Grafana across cloud infrastructure and DevOps workflows.

DevOps Intern March 2022 – July 2022
Tiger Analytics *India*

- Developed Azure DevOps CI/CD pipelines to automate infrastructure deployments on Microsoft Azure.
- Containerized a Java application using Docker and deployed it to Kubernetes.
- Created Terraform configurations for Azure infrastructure, configured virtual machines, and performed routine Linux and DevOps operations.

Machine Learning Intern April 2021 – June 2021
Indian Servers (Software Development Company) *India*

- Worked on Machine Learning workflows using Python, pandas, NumPy, Jupyter Notebook, and Google Colab, applying supervised, unsupervised, and semi-supervised learning techniques.
- Implemented machine learning algorithms including linear regression, logistic regression, decision trees, random forests, KNN, K-Means clustering, and classification.
- Developed a COVID-19 detection solution using chest X-ray images with PyTorch and transfer learning, and built a PUBG enemy detection project using OpenCV.
- Worked with ML pipelines and MLflow as part of machine learning experimentation and model development.

Projects

AKS Microservices Platform | *Azure, Kubernetes, Docker, Helm, Istio, Prometheus, Grafana*

- Designed and deployed a containerized microservices platform on AKS using Docker and Kubernetes manifests for application workloads and supporting resources.
- Configured Helm charts and Istio ingress to manage application deployments, service communication, and inbound traffic routing.
- Implemented Prometheus and Grafana monitoring for cluster infrastructure, resource utilization, request traffic, and service throughput.

Multi-Environment Azure Infrastructure Automation | *Terraform, Azure DevOps, GitLab CI/CD, Terratest, Checkov*

- Developed reusable infrastructure-as-code configurations and CI/CD pipeline templates to automate Azure resource provisioning and application deployment.
- Integrated Terratest and Checkov into infrastructure workflows for automated testing and policy scanning.
- Provisioned cloud networking, compute, storage, databases, security, and monitoring resources across multiple environments.

Certifications

Claude Certified Developer – Foundations (CCDV-F)
Certified Kubernetes Administrator (CKA)
HashiCorp Certified: Terraform Associate
AWS Certified Solutions Architect – Associate
Microsoft Certified: Azure Administrator Associate (AZ-104)
Microsoft Certified: Azure Fundamentals (AZ-900)
Python (Basic) Certification – HackerRank
5-Star Python Badge – HackerRank

Education

Adikavi Nannaya University — Rajahmundry, Andhra Pradesh
Bachelor of Technology in Electronics and Communication Engineering — *2018 – 2022*